

October 15, 2025 06:38 PM GMT

US Rates Strategy | North America

An Introduction to STRIPS

Stripping has reached \$1 trillion outstanding, with activity at 15-year highs. Fully funded pensions and strong demand for duration are driving flows, making strips increasingly important for liability matching, asset swaps, and relative value positioning.

Key Takeaways

- Strips are seeing a resurgence, with investor demand for duration pushing activity to 15-year highs and outstanding notional now around \$1 trillion.
- Pension funds have become a key driver of flows as fully-funded status and higher discount rates allow them to rotate into long-duration fixed income.
- Strips provide efficient exposure to duration and convexity across the curve, with P-strips concentrated in the long-end and C-strips in the front-end.
- Persistent pricing differences among P-strips, C-strips, Treasuries, and swaps create relative value opportunities across curves and against whole bonds.
- With duration in focus, strips are increasingly important for liability-driven investors, asset-swap participants, and those looking to express curve views.

Please [add me](#) to your distribution list.

MORGAN STANLEY & CO. LLC

Eli P Carter

Strategist

Eli.Carter@morganstanley.com

+1 212 761-4703

Matthew Hornbach

Strategist

Matthew.Hornbach@morganstanley.com

+1 212 761-1837

Aryaman Singh

Strategist

Aryaman@morganstanley.com

+1 212 761-1993

Martin W Tobias, CFA

Strategist

Martin.Tobias@morganstanley.com

+1 212 761-6076

Shaun Zhou

Strategist

Shaun.Zhou@morganstanley.com

+1 212 761-3348

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

Mind Your Ps and Cs

Why you should read this primer now

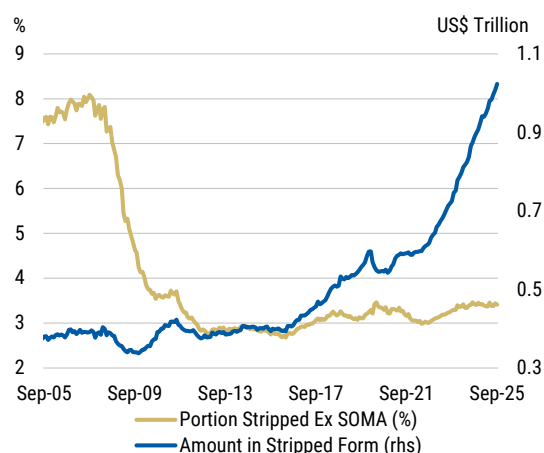
Strips are unique products in the Treasury complex. They give investors the ability to access duration and convexity cheaply, and their illiquidity premium often makes them attractive from a relative value perspective.

Interest in the product has already started to pick up over the past few years. Our business has seen increased flows for strips outright and on asset swaps, while the Treasury's monthly statement of public debt has also shown increased activity.

There are currently \$1.02tr of strips notional outstanding, a 2.6x increase in strips outstanding over the past 10 years. Despite this, the percent of eligible bonds not held in the SOMA portfolio that live in stripped form has decreased substantially over the last 20 years (see [Exhibit 1](#)) as nominal issuance began to outpace stripping after the GFC.

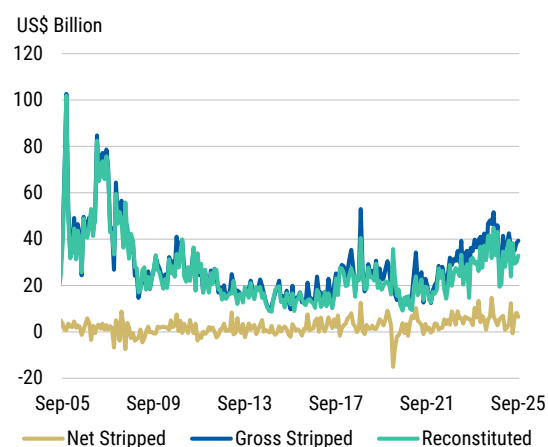
Strips have begun to make a comeback in the past few years, with the percent stripped at 15y highs. There has been increased activity in both stripping and reconstituting, and the net effect has been almost 3 years of net stripping activity (see [Exhibit 2](#)).

Exhibit 1: Notional strips outstanding outright and as a percent of eligible bonds ex-SOMA holdings



Source: Morgan Stanley Research, US Treasury

Exhibit 2: Monthly stripping and reconstitution activity

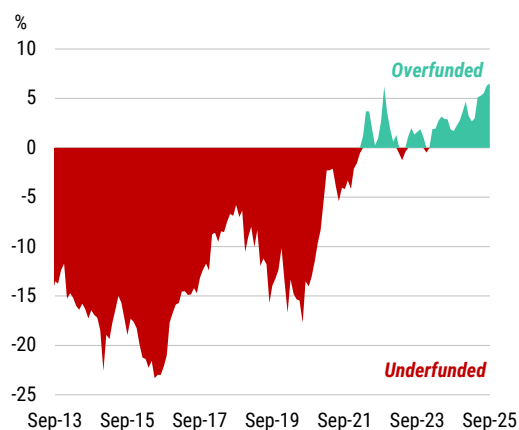


Source: Morgan Stanley Research, US Treasury

Pension funds have likely been key drivers of increased stripping activity over the past years. Pensions are a natural duration buyer given their defined future cash flows, but broadly they have been underfunded over the past 15 years (see [Exhibit 3](#)). As a result, they have tended to have a higher equity mix of investments.

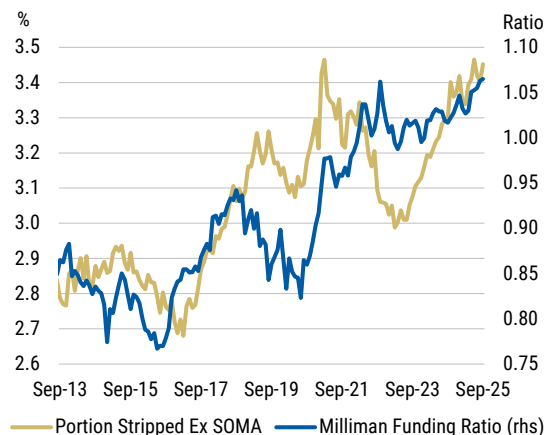
In late 2021, pensions hit and have mostly maintained their fully funded status. This has allowed pensions to rotate from equities to fixed income and better match liabilities. Strips as a percentage of non-SOMA eligible bonds started increasing around the same time, suggesting demand from these rebalancing programs, as seen in [Exhibit 4](#).

Exhibit 3: Milliman pension funding index of 100 largest defined benefit corporate pension plans



Source: Bloomberg, Morgan Stanley Research

Exhibit 4: Milliman pension funding index vs. percent of eligible bonds ex-SOMA holdings held in stripped form

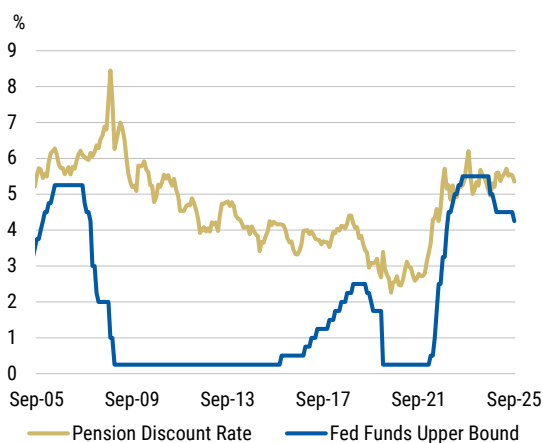


Source: Morgan Stanley Research, Bloomberg

One reason pension funds have been fully funded has been higher discount rates (see [Exhibit 5](#)). As rates have been higher, the value of their future liabilities has been lower.

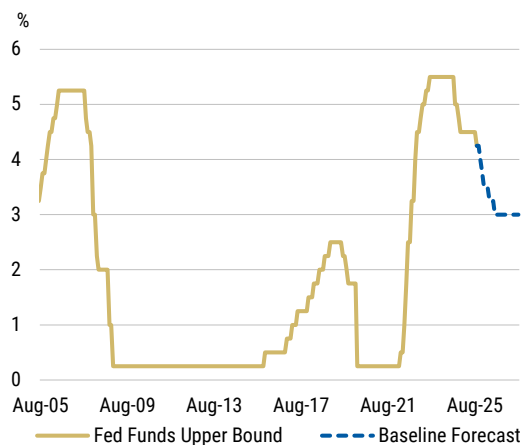
If rates were to fall, as projected by our economists (see [Exhibit 6](#)), this could pressure the funding ratios of defined benefit plans. As a result, we could see heightened urgency from pension funds to lock in cheap duration via strips in the near term.

Exhibit 5: Milliman pension funding index discount rate vs. the Fed funds rate



Source: Bloomberg, Morgan Stanley Research

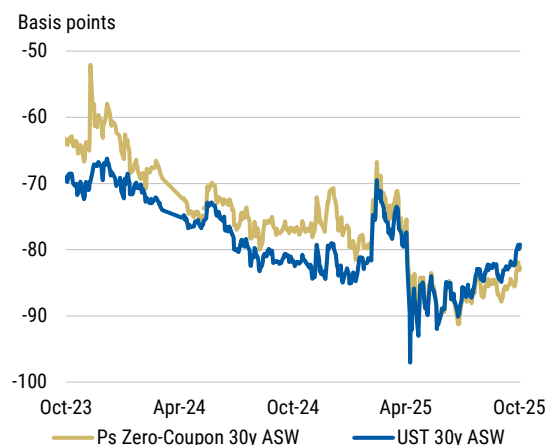
Exhibit 6: Historical and projected Fed funds rate



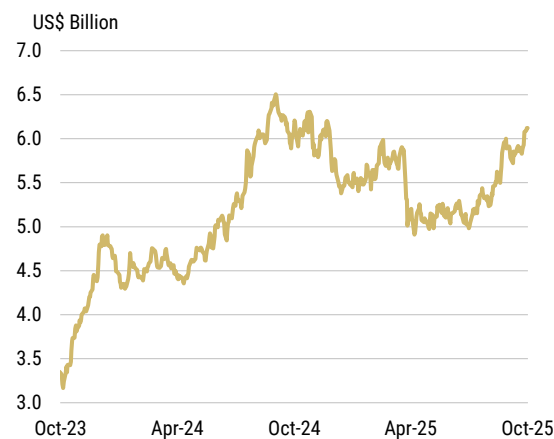
Source: Bloomberg, Morgan Stanley Research Estimates

Beyond pension dynamics, market technicals and investor flows also highlight growing demand. [Exhibit 7](#) shows that recently, the spread between long-end P-strips and their equivalent swaps has become more negative than the spread between long-end treasuries and their equivalent swaps, making buying Ps on asset swap more attractive.

Also in the last few months, the largest ETFs tracking the performance of strips has seen inflows. Since 2023, the AUM of these strips ETFs has nearly doubled and is approaching new all-time highs (see [Exhibit 8](#)).

Exhibit 7: Treasury securities spread from swap equivalent

Source: Morgan Stanley Research, Bloomberg

Exhibit 8: Total assets of top 3 ETFs tracking long-dated strips

Source: Bloomberg, Morgan Stanley Research

No investment recommendation is made with respect to any of the ETFs or mutual funds referenced herein. Investors should not rely on the information included in making investment decisions with respect to those funds.

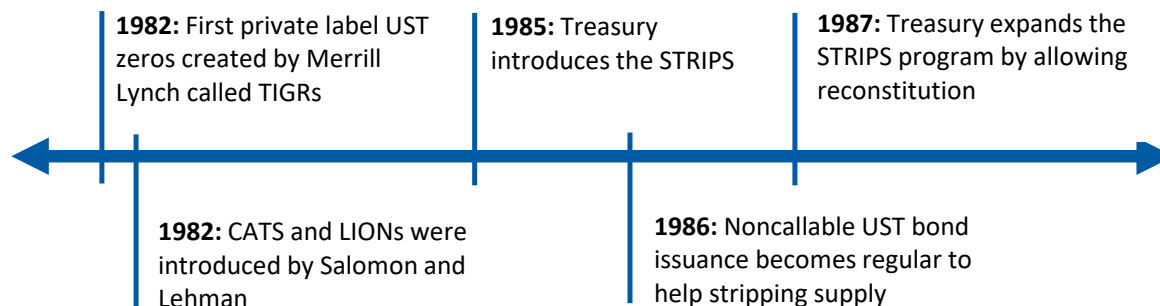
The use of strips has continually increased since 2021. As we move forward, we think understanding strips will be more and more important for investors.

What is the STRIPS program?

STRIPS is an acronym for Separate Trading of Registered Interest and Principal of Securities. Strips are zero-coupon bonds created from eligible Treasury notes' and bonds' interest (C-strips) and principal (P-strips) cash flows. Each strip entitles the owner to a single, fixed cash flow, maturing on a future date. (Note: For the purpose of this report, "STRIPS" will refer to the program, while "strips" will refer to the actual objects.)

The first zero-coupon securities based on UST cash flows were collateralized pools of UST bonds created by banks in 1982, whose structured cash flows were sold to investors.

In 1985, the Treasury created its own zeros program, recognizing the demand. Over time, strips took over the market, as they are direct liabilities of the US government with no credit risk like private-label alternatives.

Exhibit 9: A timeline of zero-coupon securities based on Treasury cash flows

Source: US Treasury, Federal Reserve Bank of Kansas City, Morgan Stanley Research

Strips, then, are the closest investors can get to a government issued zero-coupon bond, given that the Treasury does not currently issue zero coupon instruments outside bills. Before they can be invested in, they must be *created*.

How are strips made?

Strips are created by depository institutions tapping the Fed's STRIPS facility. By sending a particular order via Fedwire, the book-entry system will be changed to debit the individual CUSIP from the sender's account, and credit the CUSIPs for the corresponding strips.

When a depository institute strips a whole bond, they are taking the cash flows granted from the bond's coupons and principal and repackaging the cash flows.

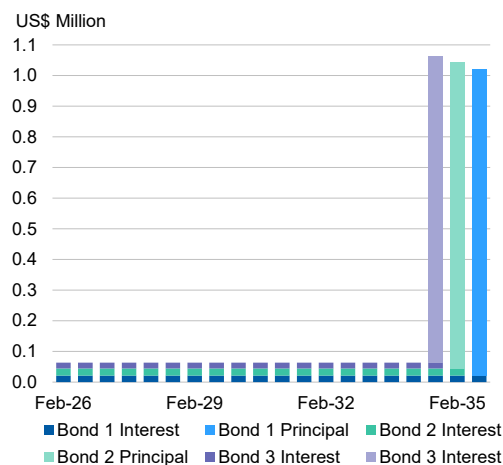
- The bond's principal is broken out and becomes its own security, a p-strip.
- The coupons are stacked with any other stripped coupons with the same payment date, becoming c-strips.

Said differently, P-strips and C-strips are both created, but C-strips are fungible and share a CUSIP with any existing C-strips that have the same maturity date.

For example, take three bonds with ~10y left until maturity: the 4.25% Aug35s, the 4.625% of Feb35s, and the 3.875% of Aug34s:

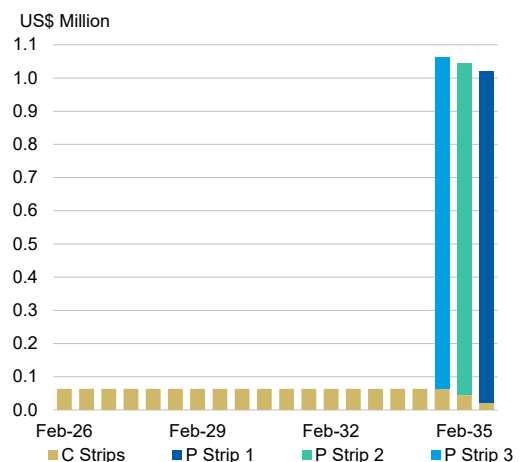
- [Exhibit 10](#) shows the cash flows from these three bonds, with there being 60 cash flows but only three instruments to trade them.
- [Exhibit 11](#) shows the repackaged cash flows after stripping. There are now 23 cash flows with 23 instruments to trade them.

Exhibit 10: Cash flows of three bonds before stripping



Source: Bloomberg, Morgan Stanley Research. Note: Cash flows are given based on a notional amount of \$1,000,000

Exhibit 11: Cash flows of three bonds after stripping



Source: Bloomberg, Morgan Stanley Research. Note: Cash flows are given based on a notional amount of \$1,000,000

The same institutions can also reverse the process – that is, to trade in all of the individual strips to the Fed and receive back the whole bond. The name for this process is reconstitution and is also done via a Fedwire instruction to the book entry system.

Outstanding stock of P- and C-strips

As noted earlier, there has been a large increase in the amount of strips over the past 10 years, with currently about \$1tr of strips notional outstanding.

[Exhibit 12](#) shows the maturity profile of the current outstanding strips universe. The distribution across P-strips and C-strips is vastly different.

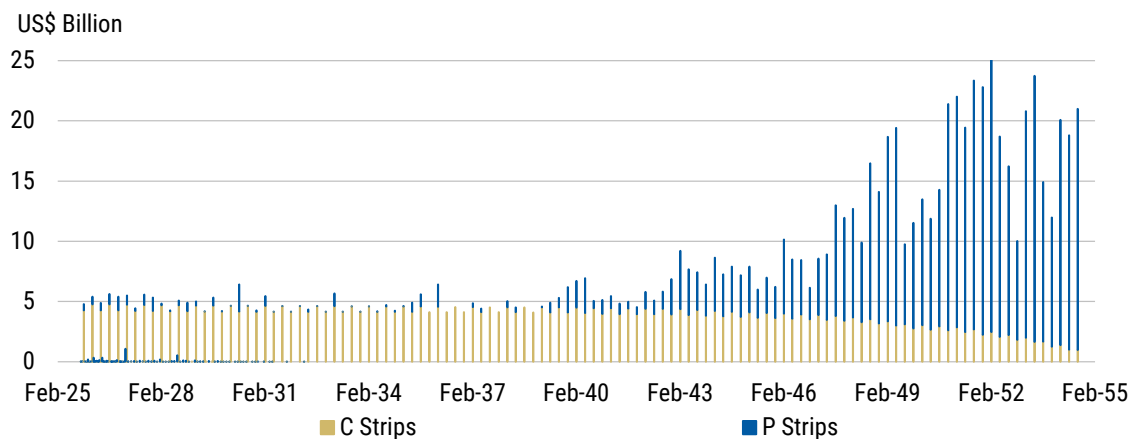
P-strips:

- Make up the majority of the long-end of the strips curve.
- Provide larger pools of liquidity for a given tenor.

C-strips:

- Available for more maturity dates, especially in the front-end of the strips curve.
- Less liquidity per tenor as they are more of a by-product of the stripping process.

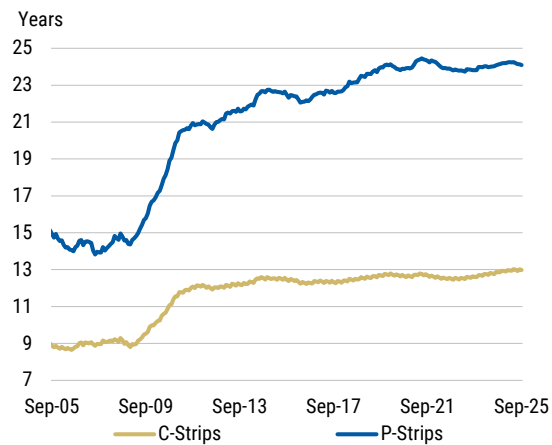
Exhibit 12: All outstanding strips as of 8/18/25



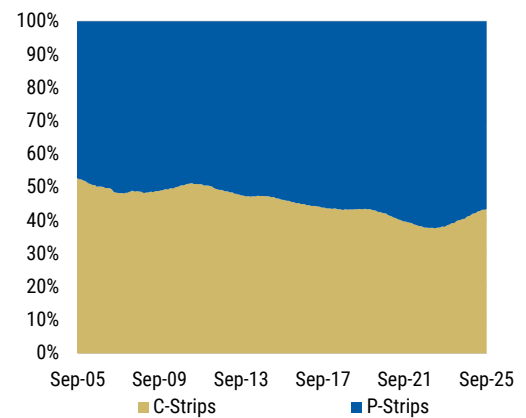
Source: Morgan Stanley Research, US Treasury

P-strips have always represented the long-end of the strips curve, as the principals mechanically come after coupons. Prior to 2009, the WAM of P-strips was ~14.5y, and for C-strips, it was ~9y. Following the WAM extension of the overall UST universe and introduction of quarterly 30y bond issuance in 2009, STRIPS WAM extended to ~24y and ~13y, respectively, as of today (see [Exhibit 13](#)).

Despite the difference in maturity profiles, P-strips and C-strips make up a similar amount of the outstanding stock of strips (see [Exhibit 14](#)). Over time, P-strips have increased their share of the outstanding as stripping bonds created from 2009-2021 produced lower amounts of C-strips given the low coupon rates.

Exhibit 13: Historical P-strip and C-strip WAM

Source: Morgan Stanley Research, US Treasury

Exhibit 14: Historical strips composition

Source: Morgan Stanley Research, US Treasury

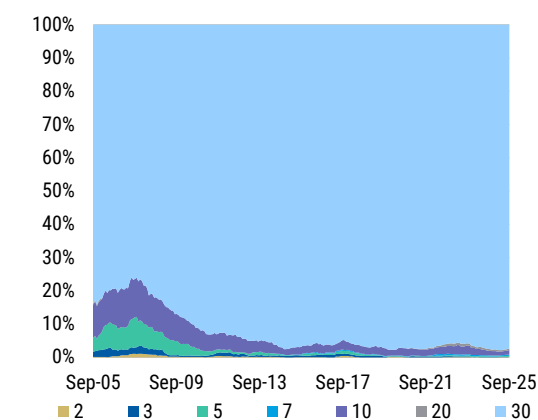
We've added summary statistics of the strips universe to our Bloomberg ticker suite:

- **MS** | Morgan Stanley
- **ST** | STRIPS
- **P, C** | Principal or coupon
- **OUT, WAM** | Outstanding or Weighted-Average Maturity

Which Treasuries are stripped and how fast?

Almost all outstanding Treasuries are eligible to be stripped, but only a subset actually are stripped in any meaningful sense (see [Exhibit 15](#)). Just 17.2% of eligible outstanding bonds have over 5% of their issuance stripped.

The most important factor in determining if a bond is likely to have a meaningful amount stripped is to look at the original maturity. 30y bonds make up almost all of the stripped bonds ([Exhibit 16](#)), as they provide attractive duration (more discussion below).

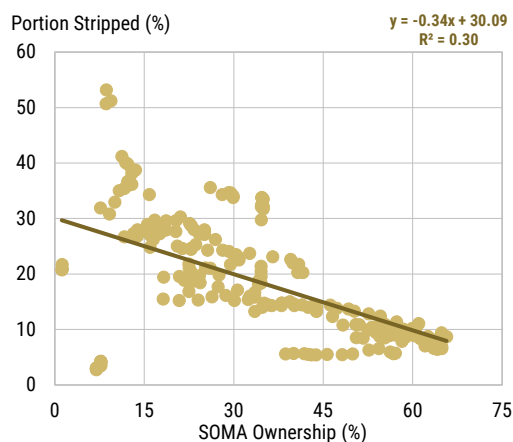
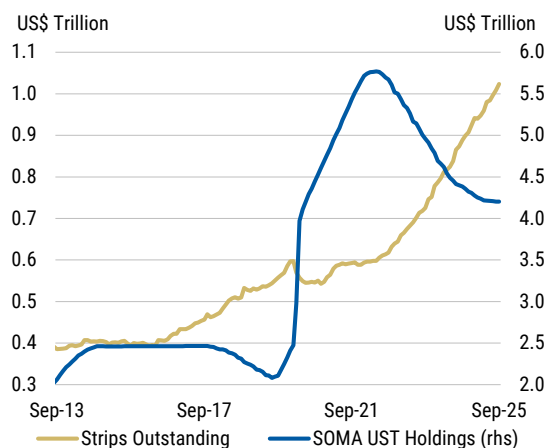
Exhibit 15: Percent of bonds with over 5% of the outstanding CUSIP stripped**Exhibit 16:** Original maturity of outstanding strips

Additionally, excess sources of demand for particular CUSIPs can make certain whole bonds more attractive to hold fully constructed. This often leads to a reduced amount being held in stripped form. This effect was evident during QE.

In [Exhibit 17](#), we look at the proportion of originally issued 30y bonds held in stripped form versus the proportion owned by the Fed's SOMA portfolio. After filtering for bonds that were near their peak stripping period (15 to 30 months after issue) and looking at QE 1, 2, and 3, we see a negative relationship.

The relationship suggests that for every additional percent of Fed ownership, the amount held in stripped form decreases by 0.35%.

While the effect on individual CUSIPs is noticeable, the overall effect of the Fed's balance sheet is not as clear ([Exhibit 18](#)). Since QT has started, strips outstanding has really taken off, but so has issuance overall and the relationship is not always clear.

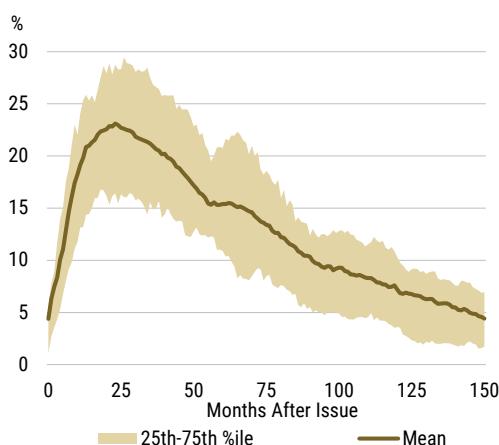
Exhibit 17: SOMA ownership's effect on portion of a CUSIP held in stripped form**Exhibit 18:** SOMA UST portfolio size and strips outstanding.

Another interesting metric to judge the popularity of stripping, both on a CUSIP and aggregate level is the rate at which bonds are stripped.

For a given 30y bond, we can track how much has been stripped as we move away from their issue date, providing a "speed-curve" pattern. As seen in [Exhibit 19](#), 30y bonds are continuously stripped for the first ~25 months of issuance, whereafter the pace slows and ends up reversing.

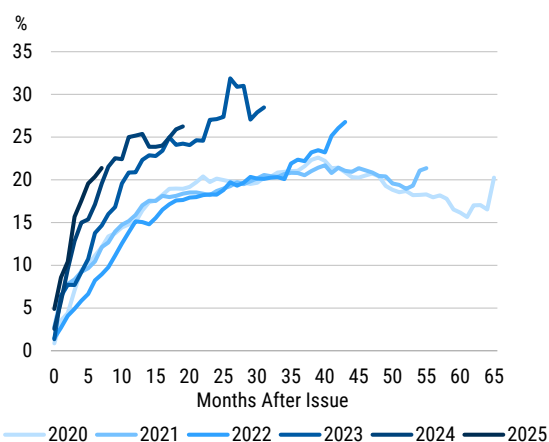
As further evidence of increased stripping activity, 30y bonds have been on increasingly faster "speed-curves" over the past years ([Exhibit 20](#)). 30y bonds are stripped more as a % of their issuance size and stripped faster.

Exhibit 19: Average percent of 30y bonds in stripped form after issue



Source: Morgan Stanley Research, US Treasury

Exhibit 20: Average stripping pattern over recent years



Source: Morgan Stanley Research, US Treasury

Properties and natural buyers of strips

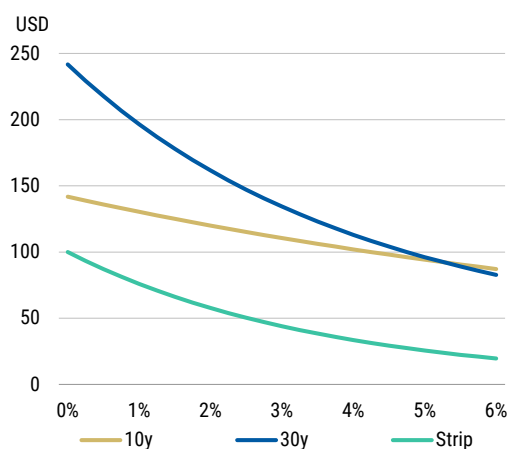
Strips have differentiated properties making them attractive to particular investors who cannot get their needs met by other Treasury products. These include:

- High duration and convexity for a low cost.
- Possible tax benefits.
- Targeted cash flows.

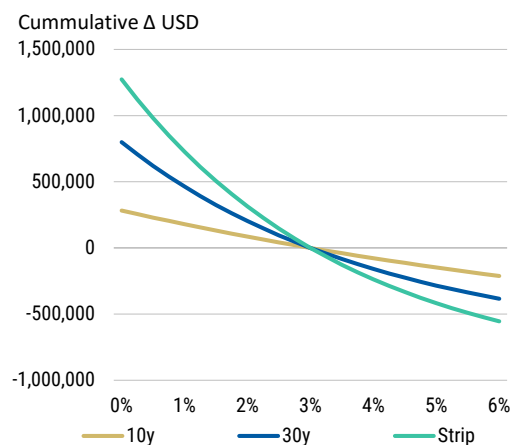
The most important differentiator of strips compared to other Treasuries is the duration risk afforded to an investor. Given that a strip is a single cash flow discounted by a factor representing a date (usually) far in the future, strips always trade below par.

[Exhibit 21](#) shows what the price, in terms of par, for SP May53, CT10, and CT30 would be based on different yields. CT30 has similar, though higher, duration than the strip, but this comes at almost twice the cost.

[Exhibit 22](#) presents the same information in a price-normalized way. When the changes in prices are visualized as relative to a set dollar investment amount, the P-strip becomes the product giving the most duration and convexity exposure.

Exhibit 21: Prices in terms of par for different yield levels

Source: Morgan Stanley Research, Bloomberg

Exhibit 22: Price changes for a million USD investment made with yields as 3%

Source: Morgan Stanley Research, Bloomberg

This high duration and convexity for a low dollar price is especially attractive for groups of investors that have long duration liabilities like pensions and insurance companies.

These long-dated liabilities also carry significant duration and convexity. Often, these can be longer in duration than those of the longest USTs. As a result, strips are important to these investors as a way to match these liabilities efficiently.

Additionally, strips often have detrimental tax structures. Despite strips being zero-coupon bonds and therefore not providing coupon income, investors must still pay annual taxes on the imputed interest.

Said differently, the amount that the bond gains in value as it approaches its face value each year is taxed as ordinary income. Thus, an investor will have a net outflow every year. This makes strips better for tax-advantaged accounts.

Lastly, strips can be used to target expenses on specific dates. If an investor has a specific date they need a cash flow, like a defeasance fund or retail investor saving for college, strips can provide some advantages.

Instead of having to deal with coupon cash flows, reinvestment, and selling the remaining bond, strips can provide exact exposure to the months desired. Liability matching can be done simply and efficiently using strips.

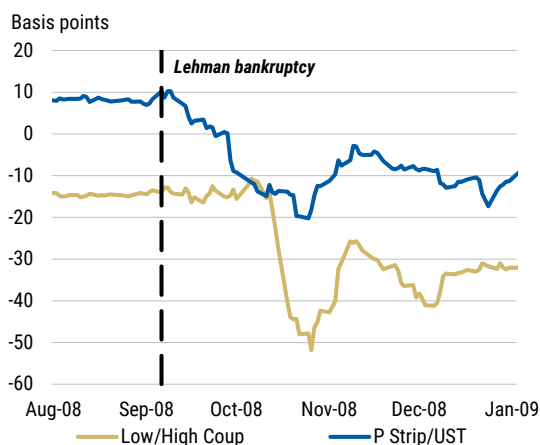
A hidden benefit of strips

Strips, being zero-coupon bonds, have low dollar prices. Most of the time, this matters little for bond investors. Usually, the yield that an investor will receive is much more important. During time of crisis however, a low dollar price can become a significant advantage.

Following Lehman's bankruptcy, in the depths of the GFC, low-coupon bonds significantly outperformed high-coupon bonds. Matching expires as best as we could, [Exhibit 23](#) and [Exhibit 24](#) show that the 4.5% May17s outperformed the 8.875% Aug17s by 40bp over the course of one month.

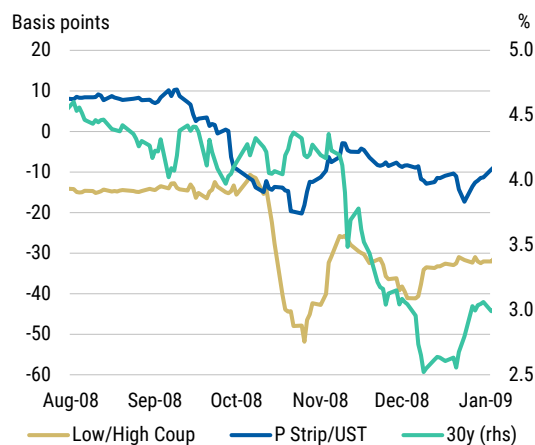
These charts also show that P-strips outperformed their whole-bond counterparts in the crisis. In the two months following Lehman's bankruptcy, the Feb36 p-strip outperformed the whole-bond by 30bp.

Exhibit 23: Low vs. high coupon USTs and P-strips vs. USTs compared to the level of rates



Source: Morgan Stanley Research, Bloomberg

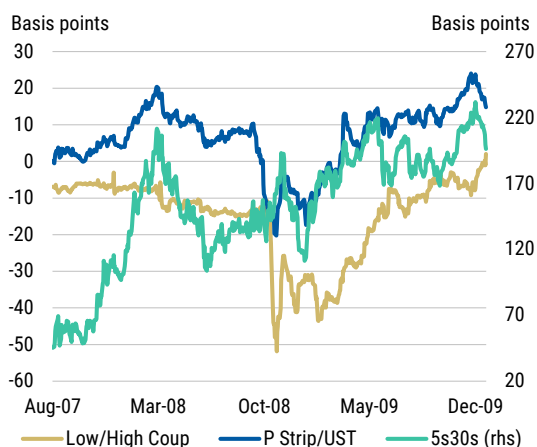
Exhibit 24: Low vs. high coupon USTs and P-strips vs. USTs compared to the level of rate



Source: Morgan Stanley Research, Bloomberg

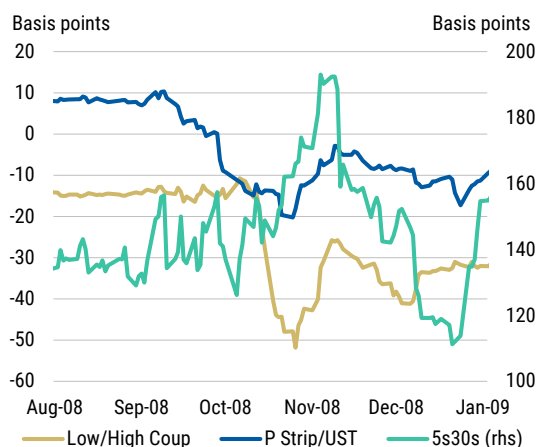
Although the low/high coupon and P/UST spread followed a similar pattern, this is unlikely an example of the coupon and curve effect (explained [here](#)). During the outperformance, the curve did not show the expected flattening (see [Exhibit 25](#) and [Exhibit 26](#)).

Exhibit 25: Low vs. high coupon USTs and P-strips vs. USTs compared to the curve



Source: Morgan Stanley Research, Bloomberg

Exhibit 26: Low vs. high coupon USTs and p-strips vs. USTs compared to the curve



Source: Morgan Stanley Research, Bloomberg

Certainly, one of the reasons that low coupons outperformed high coupons is because of the liquidity profile. The low coupons tended to be more recently issued and, as a result, were more liquid than the high coupons. However, strips are far less liquid than USTs, so this liquidity explanation doesn't explain their outperformance.

Instead, we think that the low cash prices of low coupons and especially strips compared to their equivalents (see [Exhibit 27](#) and [Exhibit 28](#)) led to their outperformance. In this time of crisis, after a major bank failed, counterparty risk and uncertainty was very high.

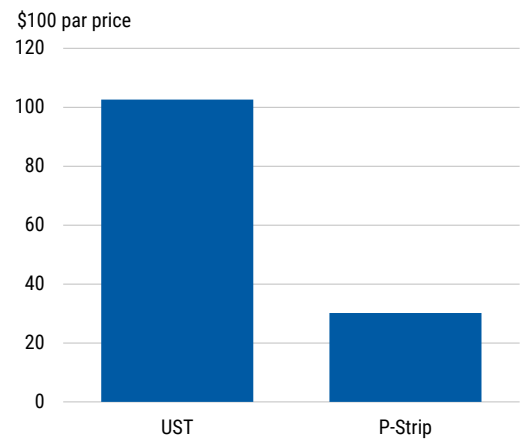
Lower cash price bonds allowed for yield exposure, while reducing the amount of money needed during settlement. This reduced some of the counterparty risk, while keeping the safe-haven status of being a US treasury.

Exhibit 27: Low vs. high coupon cash price for \$100 of par during GFC



Source: Bloomberg, Morgan Stanley Research

Exhibit 28: P-strip vs. UST cash price for \$100 of par during GFC



Source: Bloomberg, Morgan Stanley Research

Strips provide this reduced counterparty risk benefit better than any other instrument given their low cash price and that they are still liabilities of the US government.

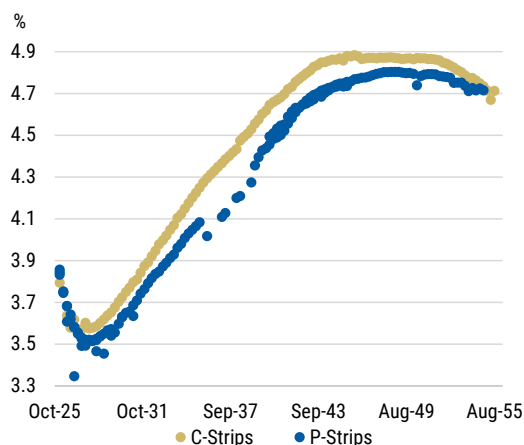
Trading Strips

Strip curves

Given that P-strips and C-strips are identical instruments, representing single cash flows from the same issuer, you may expect them to trade along a similar curve given a no arbitrage argument. If there is a P-strip and a C-strip with the same maturity, you may very reasonably assume they would have the same yield.

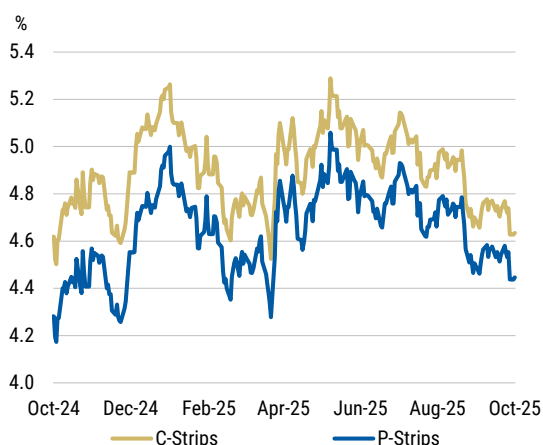
In reality, these statements are not true. Instead, when plotting strips by yield vs. maturity like in [Exhibit 29](#), two distinct curves appear, which correspond to a C-strips curve and a P-strips curve. When plotting the yield of identical cash flows with the same maturity we also find a persistent spread (see [Exhibit 30](#)).

Exhibit 29: Observed yield curves for c-strips and P-strips



Source: Morgan Stanley desk estimates, Morgan Stanley Research

Exhibit 30: Yield of two identical cash flows both maturing in May 2040



Source: Bloomberg, Morgan Stanley Research

Why does such a difference arise? The pricing of strips is a difficult problem - many spots on the curves, in particular the C-strip curve, are very illiquid. On the other hand, the C-strip curve is more "complete" than the P-strips curve, making mathematically consistent pricing easier. On top of these factors, the whole bonds that the strips were derived from actively trade and should also be a pricing input.

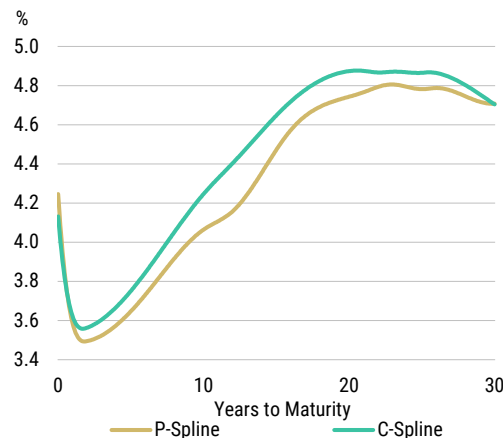
Pricing methodology varies by dealer, but often the following two steps are taken (see [Appendix](#) for more details):

- Whole bond prices are taken as given and are used as a ground truth from which you can back out P-strip prices if given P-strip prices.
- Some assumption is made on the spread between C-strips and their corresponding P-strips across the curve, allowing price calculations for C-strips. The P-strips then fault out as a result.

Therefore, most active strips market makers upkeep two strips curves, one for C-strips and another for P-strips. A simply fitted cubic spline can be seen in [Exhibit 31](#).

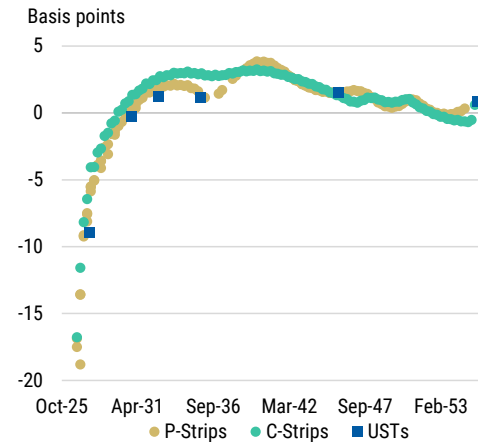
Once splines for the P-strip and C-strip curve are obtained, we can calculate the carry and rolldown for each strip. Often the curve is steeper in zero coupon space so strips can have attractive rolldown relative to USTs ([Exhibit 32](#)).

Exhibit 31: Fitted yield curves for C-strips and P-strips



Source: Morgan Stanley Research, Morgan Stanley desk estimates

Exhibit 32: 3m RD&C strips curve and CMT USTs



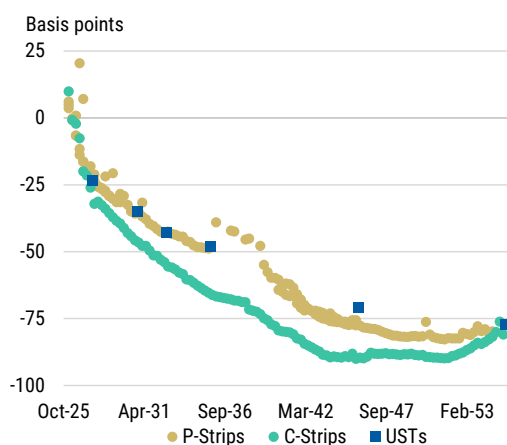
Source: Morgan Stanley Research, Morgan Stanley desk estimates

Strips are also looked at by investors on asset swaps. Similar to treasury swap spreads, investors are interested in being paid to hold the more balance sheet intensive cash instrument rather than the equivalent swap.

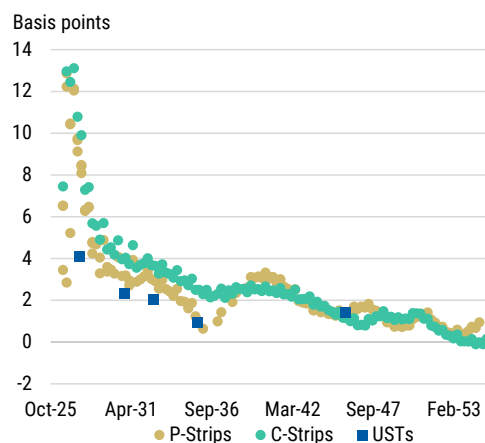
Since these are non-interest-bearing securities, the most accurate type of swap to pair with a strip trade is a zero-coupon, maturity-matched SOFR swap. Similar to strips, these do not payout until they are exited or maturity is reached.

Defining the asset swap as the yield of the swap minus the yield on the strip, [Exhibit 33](#) shows the term structure of the asset swap curve. The levels are consistently negative, and can be more so than coupon ASW.

This means that an investor could make the spread by buying a strip and paying fixed on the matched- maturity zero-coupon swap if held to maturity. Around the 10y point this would yield ~50bp running for a P-strip.

Exhibit 33: Asset swap prices across strips curve and CMT USTs

Source: Morgan Stanley Research, Morgan Stanley desk estimates

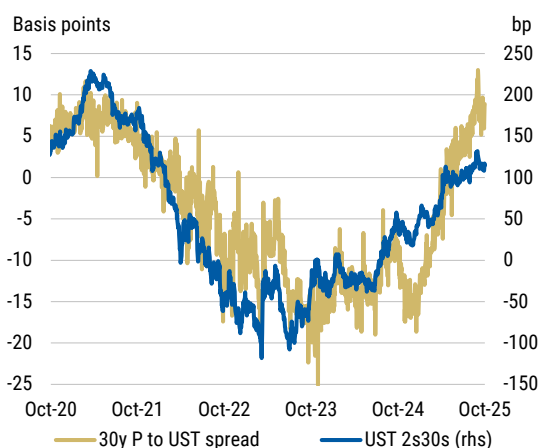
Exhibit 34: 3m RD&C for being long ASWs

Source: Morgan Stanley Research, Morgan Stanley desk estimates

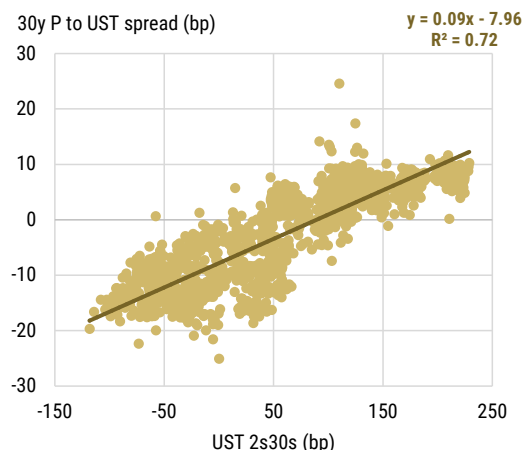
Strips and the yield curve

The spread of strips to the UST curve can be used to express curve views. As the Treasury yield curve steepens, long-end P-strips underperform an equivalent treasury, while during a flattening, they outperform the equivalent treasury (see [Exhibit 35](#)).

The relationship over the last 5 years has been strong, with an R2 of 0.72, though the coefficient is relatively small, requiring high leverage to express a curve view in this space (see [Exhibit 36](#)). This strength in relationship reduces for shorter-dated strips, and is mirrored by C-strips

Exhibit 35: Maturity matched 30y P-strips spread to UST vs. UST curve

Source: Morgan Stanley Research, Bloomberg

Exhibit 36: Maturity matched 30y P-strips spread to UST vs. UST curve

Source: Morgan Stanley Research, Bloomberg

This relationship arises in large part because of a mechanical connection. When thinking about the cash flows of a bond vs. a strip, the bond has intermittent cash flows throughout its life while the strip has only 1 cash flow at the end of its life.

The intermittent cash flows in the bond are worth more as the curve steepens. This is because they are being discounted by less, making them worth more. The strip doesn't get this benefit, and therefore underperforms in a steepening.

In a way, it's similar to how low-coupon bonds underperform high-coupon bonds in a steepening. A strip, being a zero, is just the lowest coupon bond you can get.

Bond-strips parity trade

As noted above, the pricing of C-strips, P-strips, and the whole bonds from whence they came should be related in some way. Theoretically, an investor should be indifferent between owning the whole bond versus all of the individual cash flows that make up the bond.

Said differently, the sum of the prices of the strips making up a bond should be equivalent to the price of the whole bond.

$$P(\text{Whole Bond}) = P(\text{C-Strip}_1) + P(\text{C-Strip}_2) + \dots + P(\text{C-Strip}_i) + P(\text{P-Strip})$$

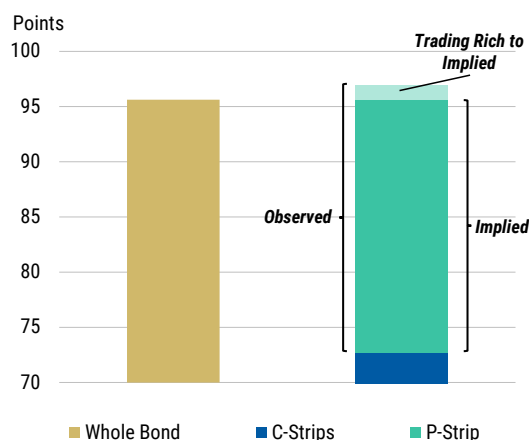
To test for that, we compare observed prices for strips and whole bonds streaming on Bloomberg. Since the whole bond is much more liquid, we take that price as a given. If we also assume the observed prices of the C-strips are correct, as the C-strip curve is easier to construct, we can back out the implied P-strip price.

Comparing the implied P-strip price to the observed P-strip price, theoretical arbitrage opportunities could arise under the following situations:

- Implied < observed: The P-strip is trading rich to its price implied by the C-strips. Sell strips and buy the bond.
- Implied > observed: The P-strip is trading cheap to its price implied by the C-strips. Buy strips and sell the bond.
- Implied = observed: The P-strip is trading at its price implied by the C-strips.

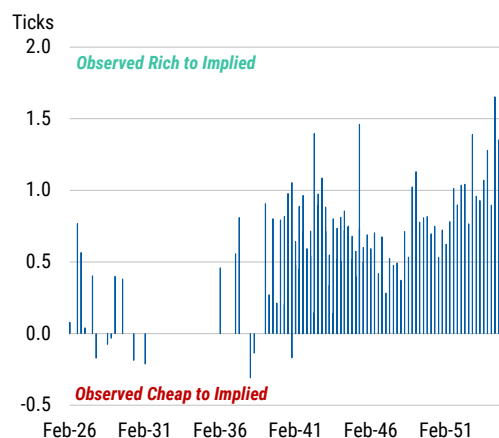
For example, [Exhibit 37](#) shows the price of the cash flows making up the May55s on October 10 in both whole bond and strips form. The observed price of the P-strip is larger than the implied, indicating that the strips are rich compared to the whole bond.

Exhibit 37: Price of the 4.75% May55s cash flows in whole bond form and strips form



Source: Morgan Stanley Research, Bloomberg

Exhibit 38: P-strips trading rich/cheap to implied levels across the curve

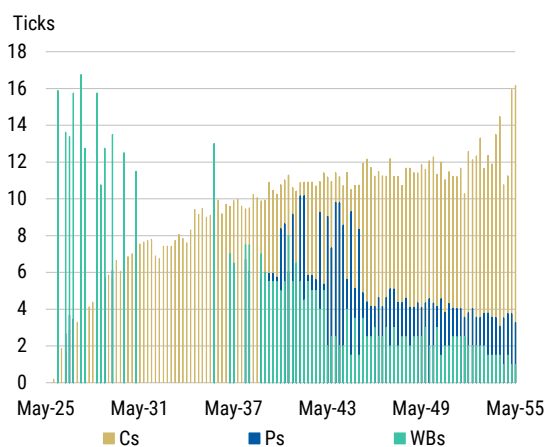


Source: Morgan Stanley Research, Bloomberg

We note that Bloomberg prices tend to show P-strips trading rich to implied prices (see [Exhibit 38](#)). If we were able to trade at mid across all the strips that we needed, a profit could be locked in. Unfortunately, these arbitrage opportunities almost always disappear once accounting for bid-ask spreads on separate, and sometimes illiquid, CUSIPS.

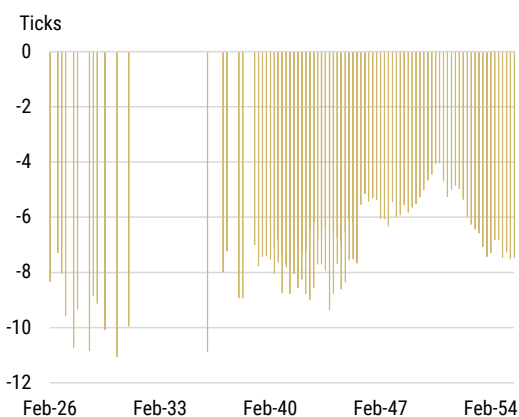
[Exhibit 39](#) shows the bid-ask spread across the strips and whole bonds curves. Trying the reconstitution arbitrage trades accounting for the spread results in a 4-to-12 tick loss across each wb/strips group (see [Exhibit 40](#)).

Exhibit 39: Bid-ask spread across strips and whole bonds



Source: Bloomberg, Morgan Stanley Research

Exhibit 40: Reconstitution trade PnL after crossing bid-ask spreads



Source: Bloomberg Morgan Stanley Research

Due to this, outright arbitrage trades are not possible – however, this type of analysis allows possible investors to pick one strip over another based on relative rich or cheapness. They can identify particular points on the curve that offer relatively more value than others.

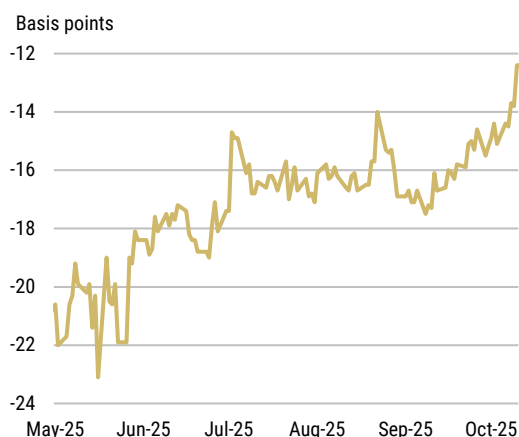
Appendix

Pricing STRIPS

Special thanks to the Xavier Zambrana and the Morgan Stanley Trading Desk for guidance on this section.

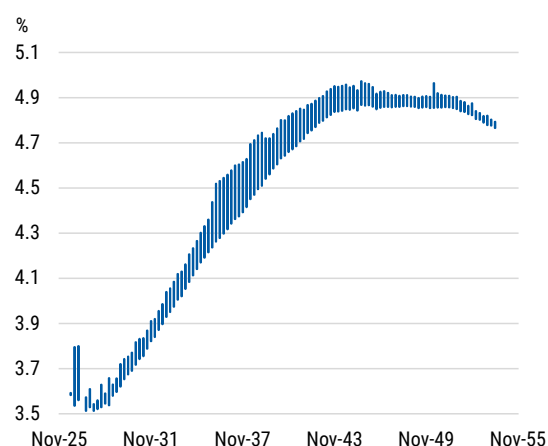
Strips seem as though they should be easy to price. They are zero-coupon bonds after all, which should give the Treasuries space a clean spot curve from which to price all other bonds. Instead, strips' illiquidity causes contradictory cash flow pricing, with bonds that represent the *exact* same cash flows having different prices (see [Exhibit 41](#)).

Exhibit 41: 50:50 butterfly between two separate principal strips and a coupon strip that expire in August 2044



Source: Bloomberg, Morgan Stanley Research

Exhibit 42: Example C-P spread separation across the curve



Source: Bloomberg, Morgan Stanley Research

To price strips while avoiding these issues, many dealers upkeep two separate curves for strips. More precisely, they often upkeep a C-strip curve, and a C-P spread curve (see [Exhibit 42](#)). Each dealer's spread curve ends up being slightly different, so true "mid" can be hard to find.

Let's see how to price the strips, given a C-P spread curve and taking the whole bond pricing as ground truth. First, we start with the definition of what a bond is, the present value of a series of cash flows (1):

$$PV_{WB} = \sum_{i=1}^n PV_{C_i} + PV_P \quad (1)$$

The present values of the components can be thought of as the equations below, with the price (PX) of each component, the coupon of the WB given, and the accrued interest (AI) given as \$100 of par:

$$PV_{WB} = PX_{WB} + AI$$

$$PV_{C_i} = \frac{PX_{C_i}}{100} * \frac{Coupon_{WB}}{2}$$

$$PV_P = PX_P$$

Plugging back into (1):

$$PX_{WB} + AI = \sum_{i=1}^n \frac{PX_{C_i}}{100} * \frac{Coupon_{WB}}{2} + PX_P \quad (2)$$

Starting with simplest case, we look at a bond with one coupon payment left:

$$PX_{WB} + AI = \frac{PX_{C_1}}{100} * \frac{Coupon_{WB}}{2} + PX_P$$

Since we assumed the price of the WB is a given, this leaves us with two unknowns, the price of the coupon cash flow and the price of the principal cash flow. Since these cash flows are economically identical, but have different supply/demand dynamics, dealers upkeep a C/P spread curve:

$$PX_P = PX_{C_n} + Spread_{price_n}$$

$$Spread_{price_n} = Spread_{yield_n} * \frac{\frac{dp}{dy_P} + \frac{dp}{dy_{C_n}}}{2}$$

Using this, we will then only have 1 unknown in our simple case, the price of the C, which we can solve for with everything else known:

$$PX_{WB} + AI = \frac{PX_{C_1}}{100} * \frac{Coupon_{WB}}{2} + PX_{C_1} + Spread_{yield_1} * \frac{\frac{dp}{dy_P} + \frac{dp}{dy_{C_1}}}{2}$$

$$(PX_{WB} + AI - Spread_{yield_1} * \frac{\frac{dp}{dy_P} + \frac{dp}{dy_{C_1}}}{2}) (1 + \frac{Coupon_{WB}}{2 * 100})^{-1} = PX_{C_1}$$

Stepping out the curve, the next WB has two coupon payments left. Here we show the set up for this bond from calculation (1) and the simplified expression solved for the price of the second coupon, given that we already found the first coupon's price above:

$$PX_{WB} + AI = \frac{PX_{C_1}}{100} * \frac{Coupon_{WB}}{2} + \frac{PX_{C_2}}{100} * \frac{Coupon_{WB}}{2} + PX_P$$

$$(PX_{WB} + AI - \frac{PX_{C_1}}{100} * \frac{Coupon_{WB}}{2} - Spread_{yield_2} * \frac{\frac{dp}{dy_P} + \frac{dp}{dy_{C_2}}}{2})(1 + \frac{Coupon_{WB}}{2 * 100})^{-1} = PX_{C_2}$$

Generally, assuming all the previous coupon prices have been calculated, and there is a C-P spread, the rest of the coupon curve can be filled out via calculation (3):

$$(PX_{WB} + AI - \frac{Coupon_{WB}}{2 * 100} * \sum_{i=1}^{n-1} PX_{C_i} - Spread_{yield_n} * \frac{\frac{dp}{dy_P} + \frac{dp}{dy_{C_n}}}{2})(1 + \frac{Coupon_{WB}}{2 * 100})^{-1} = PX_{C_n} \quad (3)$$

(3) allows you to price all of the c-strips and, by adding the C-P spread, gives you the theoretical P-strip prices.

The math really isn't the tricky part of pricing consistent strip curves. The difficulty is coming up with the C-P spread curve and the upkeep of it as yields change and the approaches to do vary by trading desk.

Source for all calculation exhibits: Morgan Stanley Research

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley & Co. LLC, and/or Morgan Stanley C.T.V.M. S.A., and/or Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., and/or Morgan Stanley Canada Limited. As used in this disclosure section, "Morgan Stanley" includes Morgan Stanley & Co. LLC, Morgan Stanley C.T.V.M. S.A., Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., Morgan Stanley Canada Limited and their affiliates as necessary.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/researchdisclosures, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Eli P Carter; Matthew Hornbach; Aryaman Singh; Martin W Tobias, CFA; Shaun Zhou.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from United States of America.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: United States of America.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2025)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1499	41%	389	46%	26%	702	41%
Equal-weight/Hold	1618	44%	375	44%	23%	782	45%
Not-Rated/Hold	4	0%	1	0%	25%	1	0%
Underweight/Sell	577	16%	88	10%	15%	234	14%
Total	3,698		853			1719	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding the relationship between the companies that are the subject of Morgan Stanley Research and Morgan Stanley Smith Barney LLC or Morgan Stanley or any of their affiliates, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to www.morganstanley.com/researchdisclosures.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf)

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or

derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 14-9169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

The following authors are Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity securities: Aryaman Singh; Shaun Zhou; Eli P Carter; Matthew Hornbach; Martin W Tobias, CFA.

© 2025 Morgan Stanley